

An Adaptive Backward Control Battery Equalization System for Serially Connected Lithium-ion Battery Packs

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Abstract—This paper presents an adaptive controller for battery equalization system for serially connected lithium-ion battery packs. The proposed equalization scheme consists of software and hardware parts to implement the adaptive neuro-fuzzy algorithm. The proposed software and hardware combined implementation of the adaptive neuro-fuzzy algorithm provides off-line learning ability for tracking dynamic reactions on battery packs, and high-speed response for equalizing currents in the individual cell equalizers. The output currents driving Pulse-Width-Modulation (PWM) signals are generated from the proposed hardware analog controllers. A feedback line is utilized for observing these output currents for the training process. The adaptive neuro-fuzzy algorithm is implemented in the main processor providing adaptive parameters for the hardware. The proposed battery equalization system has adaptability and tracking ability to deal with dynamic reactions of series connected battery cells. The hardware controllers are implemented in a 0.13 μ m CMOS technology with supply voltage of 2.5V. The results demonstrate the proposed scheme has the ability to learn previous stages, and provide precise model of the battery cell voltages and currents. The proposed system achieved learning accuracy error of 1.8×10^{-5} .

Index Terms—Battery equalization, cell balancing, analog neuro-fuzzy control, adaptive method, DC-DC converter.

I. INTRODUCTION

IN battery system, Li-ion cells are more and more replacing the traditional acid batteries, due to its lightweight and higher energy density. They are widely used in products such as the portable consumer electronic devices. Each lithium-ion battery cell can produce voltage from 1.5V to 4.5V. However, applications such as electronic vehicles (EV) require much higher voltages. Such high voltage can be achieved through serially connected cells. Generally, an EV battery pack consists of hundreds of connected individual cells. However, the imbalance between cells is a severe problem in these high voltage battery systems. Since in serially connected battery system, even though the cells are charged and discharge as the same, the small imbalance between cells cause the different reactions. This imbalance is exacerbated during working time. After number of charge and discharge operations continuously, there are weaker cells and stronger cells in a battery string [1].

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The weaker cell is the cell which could be easy to discharge and takes longer time to be fully charged than the others, and vice versa to the stronger cell. Therefore, weaker cells would usually witness a phenomenon called in-depth discharge, while the stronger ones would be subjected to overcharging. In other hand, this imbalance causes the derivation of battery cell voltages. Consequently, the imbalance battery cell voltages exacerbate the physical imbalance cells. These phenomena cause premature failures of the cells, and possible damage to the cells. Therefore, in serially connected battery string, balancing voltages between the battery cells is essential for cell protection and prolongation of the battery life. A number of research activities have been published about battery cell equalizers [1]–[17]. Equalization methods can be categorized into two groups: energy-consuming and energy-recovery or dissipative and non-dissipative methods, respectively. In dissipative methods, the unbalanced energy of each battery is converted to heat by a resistive shunt circuitry. The current dissipated by the resistive shunt is proportional to the battery cell voltage. As the result, charge balancing can be achieved. This method has advantages derived from a simple control, stability and easy implementation [2], [3]. However, the excess energy from higher voltage cells is transferred to heat that causes the rising temperature inside battery strings. Continuous working on high-temperature condition could shorten battery life and high risk of explosion [2]. Since the aging processes in the positive and negative electrodes of battery cell are strongly depend on operating temperature. In addition, voltage monitors and thermal monitors should be attached to each cell in this kind of system [4]–[6]. Energy recovery is the solution to overcome the disadvantage of additional losses in the previous methods. In the recent times, integrated individual cell equalizer (ICE) based on DC-DC converter is under development for a serially connected battery pack, such that implementation and individual control of equalization are easily achieved [1], [5]–[17]. The battery system that uses ICE scheme is shown in 1. However, the complex control of DC-DC converters for a large number of cells is one of the problems. Moreover, the higher power density requires the higher performance of DC-DC converter. Therefore, in the ICE based equalization scheme, the control circuit is one important part which contributes to the performance of battery equalization.

The equalization control restricts battery cells operation range to prevent over-discharging and over-charging, which cause subsequent damage to the active materials in the battery such

as electrodes. Depending on the design and chemistry of lithium cell, there are different nominal voltages in commerce. For example, most of lithium battery has maximum voltage below 4.2V, and minimum voltage about 2.5V. During charging state, the battery voltage should be limited to the maximum value. Otherwise, the internal pressure could accumulate and the temperature could rise to an extremely high value, which may lead to high risk of explosion. On the other hand, over-discharging will dissolve the copper in the electrolyte resulting in shortened battery life. Hence, discharging state of battery should not drop below its low voltage threshold. The controller is designed to obtain maximum usable capacity from battery cells in order to achieve high efficiency and reduce equalization time. Serially connected battery cell system is different with parallel connected system which all the cell have same voltage during the working operation, the voltage cells in the serially battery string could be different. Therefore, the voltage imbalance exacerbates the cell mismatching which caused by manufacturing variance in the physical volume available to store charge [18]. The imbalance problem is prone to exacerbation while the battery is in use. Hence, balancing the lithium-ion cells with a cell-balancing controller is necessary. For desirable result, a low response time and adaptive cell-balancing algorithm is requested. Meanwhile, previous control methods [10]–[12] are based on imbalanced state-of-charge (SOC) between cells, and the gradients of ambient temperature of the battery pack during charging and discharging processes. Recently, Yuang-Shung Lee et al [13] proposed a control method to handle equalization current between two neighboring battery cells based on fuzzy logic theory. Although fuzzy logic has advantages to model non-linear system, the previous control methods based on fuzzy logic [10]–[13] require the knowledge and experiences of the battery cells. In a prolonged ongoing system, cell voltages in battery pack change continuously and non-linearly depending on both objective and subjective parameters, such as temperature, workload etc. As a result, these conventional methods are short of adaptability to the dynamic system.

To overcome these weaknesses of previous topologies, an adaptive backward control battery equalization system is proposed focusing on enhancing control method. The cell voltages are controlled by the driving pulse-width modulation (PWM) signals, based on the equalization algorithm scheme in the analog adaptive neuro-fuzzy controllers (AANF), which are called the cell controllers. An adaptive neuro-fuzzy control scheme is constructed in the proposed system that is implemented in a 0.13- μ m CMOS technology with power supply of 2.5V. This adaptive neuro-fuzzy inference algorithm is also constructed in the Battery Management System (BMS) microprocessor, providing off-line training, system tracking and estimation of the proposed BMS scheme.

The advantage of the proposed system is inheriting adaptability of back propagation neuron network for training and estimating that are suitable with dynamic system like serially connected battery cells. Additionally, the proposed system provides the hardware circuit of controllers which are built up with ICE, so that it enables quick responds to adapt with the internal changes in individual cells. The circuit parameters

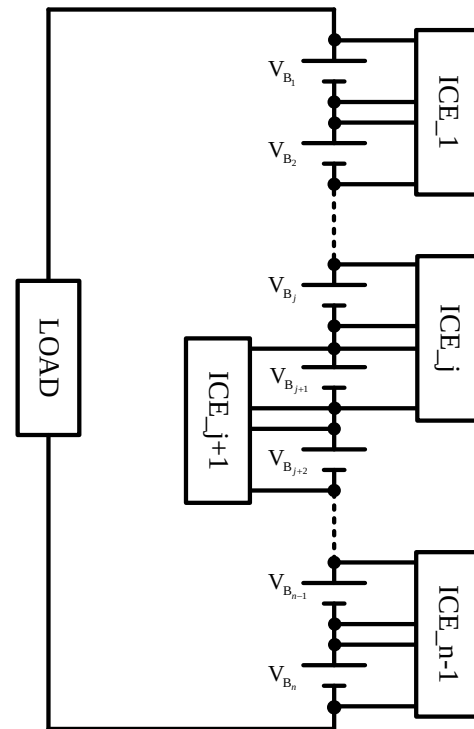


Fig. 1. Serially connected battery system using individual cell equalizers (ICEs).

of the cell controllers could be altered by communicating with the BMS microprocessor. The training and estimating process, which are time-consuming processes implemented in the main controller, could not affect the equalization time in individual battery equalization. In section II, general architecture of an adaptive neural fuzzy inference system (ANFIS) and its application to the proposed system is introduced. The proposed cell controller is described in section III. To demonstrate the efficiency and adaptation of the proposed method, the system performance results are conducted in section IV to evaluate system performances. Section V is the conclusion.

II. APPLICATION OF ANFIS MODEL ARCHITECTURE

Functionally, the ANFIS architecture is equivalent to the first order Takagi-Sugeno fuzzy system [19]–[21]. This algorithm is a hybrid system of combination of fuzzy logic theory and neural network architecture. The use of fuzzy logic allows describing the behavior of a complex nonlinear system by simple if-then rules. This provides an easier solution of a given problem that is difficult to having accuracy models. However, in many applications, the knowledge describing the expected behavior of the system contains cluster data, which are difficult to model the relations of inputs and outputs. In these cases, the designer is required to elaborate if-then rules carefully. For the static system, the disadvantage of fuzzy logic could be overcome by a tremendous effort of the designer. However, it exacerbates the dynamic systems that are delicately handled.

While fuzzy logic provides inference mechanism under cognitive uncertainty, computational neural network offers

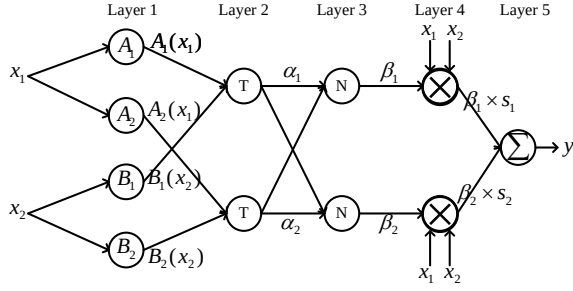


Fig. 2. ANFIS architecture of TS fuzzy control reasoning method.

exciting advantages such as learning and adaptation. The resulting hybrid system of combining fuzzy logic and neural network is called an adaptive fuzzy network. The ANFIS system is one of these hybrid systems. In an adaptive network, the network structure consists of nodes and directional links through the nodes. Moreover, the parameters in these nodes are adaptive, meaning the outputs depending on these parameters are adaptive. A learning rule specifies how these parameters should be changed to minimize a prescribed error measure.

There is combination of the least-squares method and back-propagation gradient descent method for training an adaptive networks parameters [19]. In other words, a training data set is required that contains desired input/output data pairs of the target system to be modeled. In the proposed system, the ANFIS is applied and implemented in a micro-processor so that the learning process does not affect equalizing time. The ANFIS structure used in fuzzy controller consists of five layers as shown in 2. For simplicity, only two rules with two linguistic terms for each input are considered. The following part will present the layer structure of an ANFSIS system.

- Layer 1: the output of the node is the degree to which the given input satisfies the linguistic label associated to this node. The bell-shape membership function is chosen for simplicity of analog circuit design.

$$A_i(x_1) = \left[1 + \left(\frac{x_1 - a_{i1}}{d_{i1}} \right)^2 \right]^{-1} \quad (1)$$

$$B_i(x_2) = \left[1 + \left(\frac{x_2 - a_{i2}}{d_{i2}} \right)^2 \right]^{-1} \quad (2)$$

Where: $\{a_i, d_i\}$ is the *premise parameter* while A_i, B_i are linguistic labels.

- Layer 2: each node computes the firing strength of the associated rules. The outputs of these neurons are labeled by T because of choosing $T - Norm$ for modeling, and are calculated as following:

$$\alpha_1 = A_1(x_1) \times B_1(x_2) = A_1(x_1) \wedge B_1(x_2) \quad (3a)$$

$$\alpha_2 = A_2(x_1) \times B_2(x_2) = A_2(x_1) \wedge B_2(x_2) \quad (3b)$$

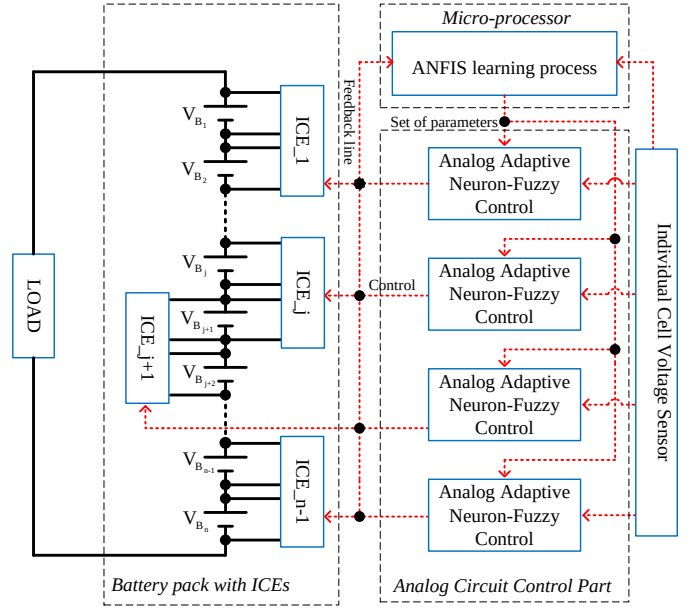


Fig. 3. The framework of adaptive backward battery equalization system.

- Layer 3: every node in this layer is labeled by N to indicate the normalization of firing levels.

$$\beta_1 = \frac{\alpha_1}{\alpha_1 + \alpha_2}, \beta_2 = \frac{\alpha_2}{\alpha_1 + \alpha_2} \quad (4)$$

- Layer 4: the outputs of these nodes at this layer are the product of the normalized firing level and the individual rule output of associated rule.

$$\beta_i s_i = \beta_i (p_i x_1 + q_i x_2) \quad (5)$$

Where $\{p_i, q_i\}$ are the set of *consequent parameters* associated the i -rule.

- Layer 5: the single output of this layer computes the overall system output as the sum of all the incoming signals.

$$y = \sum_i \beta_i s_i \quad (6)$$

The above layered structure is viewed as neural network. This structure can adapt its antecedent and consequent parameters to improve performance of the system by observing outputs. By employing these advantages of ANFIS architecture, the proposed battery equalization system has ability of learning and adaptation to deal with dynamic changes as described in the following sections.

III. PROPOSED ADAPTIVE BACKWARD CONTROL BATTERY EQUALIZATION SYSTEM

3 shows the framework of the proposed battery equalization system that is based on individual cell equalizer (ICE) that balances each pair of the neighboring cells. In the proposed system, a complete cell voltage equalizer is performed using a bi-directional dc-dc converter modified from the $C\hat{u}k$ converter [8]. The balancing currents are controlled by driving a pulse-width-modulation (PWM) signal corresponding to the

control signals constructed in the analog adaptive neuron-fuzzy control parts. In conventional schemes, it is difficult to handle the battery cell model for describing the equalizing characteristic of the lithium-ion battery strings due to electrochemical reactions and ambient temperature. Fuzzy logic control could be employed to model this non-linear and dynamic equalization during charging and discharging [12]–[14]. However, the limitation of fuzzy system is that it cannot automatically acquire the rules to make output decisions. In addition, it is challenging to model the membership functions in small change of voltage offsets with conventional fuzzy-based control methods that cause degradation accuracy of the system employing such control methods. In this work, combining neuron network in fuzzy logic control is used to tune membership functions of fuzzy system, which enables adaptability and learning ability. In the proposed system, a feedback line is provided to allow updating and tracking the changes of battery strings. Generally, the learning process of neuron network is time-consuming process. Therefore, the proposed system architecture designs off-line learning process based on a micro-processor. The analog adaptive neuron-fuzzy controllers are design by analog circuitry method to allow high-speed responses that improve equalizing speed [22]. The proposed adaptive backward control battery equalization system (ABC-BES) is a hybrid system that can deal with a complex situation and still achieve high speed in equalizing time. The following parts will elaborate the proposed ABC-BES.

A. Equalizing Circuit

Among energy converter balancing methods, the equalization methods using a centralized flyback converter is charge type and move the charge by flyback of the forward operation using a multi-winding transformer. It requires a small number of active switches but require the same number of multi-windings as the number of cells. Recently, the bidirectional flyback converter with dc-link capacitor in [16] is the direct cell-to-cell method, which is one of the charge-discharge types and the charge moves between through a dc-link capacitor by flyback operations. However this scheme requires as many transformers as the same number of cells and additional dc-link capacitors. A single magnetic cell-to-cell charge equalization converter in [1] reduced the number of transformer windings. These aforementioned methods using flyback converter has advantages of easy implementation, however, numerous magnetic devices cause significant the magnetic losses. Some other equalization schemes base on resonant converter. The scheme on [22] required the double number of inductors and switches in compared with the number of cells. The advantages of these topologies come from reducing the switching losses thus increasing the balancing system efficiency. It drawback is derived from very complex control as well as high converter cost, especially when the number of cell increase. In the proposed system, the $C\hat{u}k$ DC-DC converter topology is employed, because of its simple circuitry and small size without transformers. The enhancement for optimal control method is developed to increase the DC-DC converter efficiency while

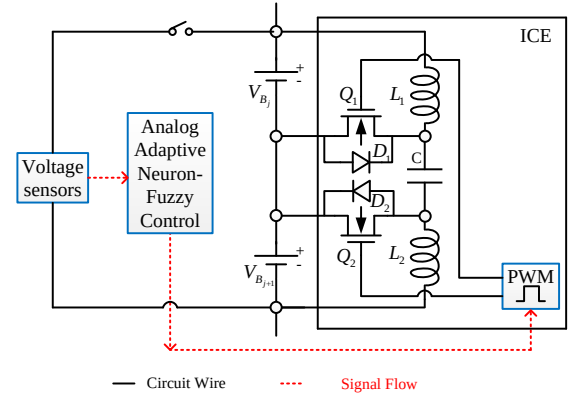


Fig. 4. Individual cell equalization circuit with the proposed controller.

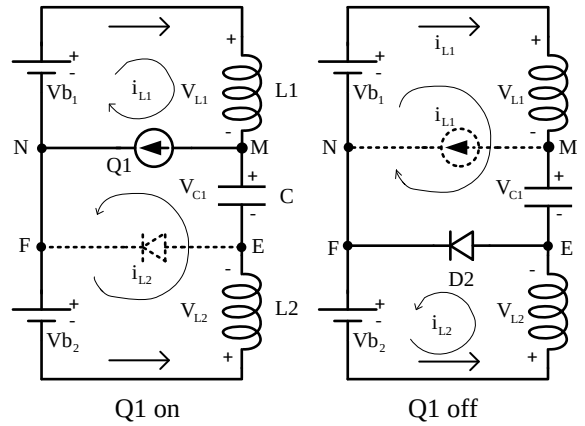


Fig. 5. Cell equalization circuit analysis.

the number of cell grows up. The equalizing circuit works on a cell-to-cell basis, therefore only one ICE_1 is analyzed for the proposed equalization as illustrated in 4. V_{B1} and V_{B2} denote the voltages of battery cell 1 and cell 2, respectively. Control signal from PWM turns on or off MOSFET Q_1 or Q_2 with control frequency f_s of or period of $T_s = 1/f_s$. Assuming $V_{B1} > V_{B2}$, battery cell 1 will be discharged and battery cell 2 will be charged. At the initial state, the voltage of the capacitor C equals to $(V_{B1} + V_{B2})$. During the first period (T_1) while Q_1 is turned on, the energy stored in the capacitor transfers to cell 2 and the inductor L_1 stores energy transferred from V_{B1} of cell 1. Likewise, the inductor L_2 also stores energy. In the remaining period (T_2) of the PWM control signal, Q_1 is turned off, and D_2 is forced to turn on. Hence, the capacitor is charged by energy stored in L_1 , and the energy stored in L_2 continues to charge cell 2. The equivalent circuit for the equalization scheme is shown in 5. The energy transformation is continuous until the circuit reaches stable state. The quantitative analysis for the equivalent circuit in 5 can be expressed with the following equations:

When Q_1 is turned on: ($0 \leq t < T_1$)

$$\begin{cases} V_{B1} = L_1 \frac{di_{L1}}{dt} \\ V_{B2} = -L_2 \frac{di_{L2}}{dt} + \frac{1}{C} \int i_{L2} dt \end{cases} \quad (7)$$

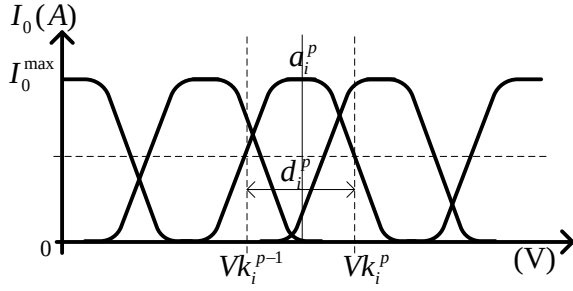


Fig. 6. Node function of neurons in layer 1 of the proposed ABC-BES.

When Q_1 is turned off: ($T_1 < t \leq T_s$)

$$\begin{cases} V_{B1} = L_1 \frac{di_{L1}}{dt} + \frac{1}{C} \int i_{L1} dt \\ V_{B2} = -L_2 \frac{di_{L2}}{dt} \end{cases} \quad (8)$$

Combining with the initial conditions and limitations, the average of inductor currents through L_1 and L_2 is computed as followed:

$$\begin{cases} I_{L1} = \frac{1}{2} T_s \left(\frac{V_{B1}}{L_1} T_1^2 + \frac{V_C - V_{B1}}{L_1} T_2^2 \right) \\ I_{L2} = \frac{1}{2} T_s \left(\frac{V_{B2}}{L_2} T_2^2 + \frac{V_C - V_{B2}}{L_2} T_1^2 \right) \end{cases} \quad (T_2 = T_s - T_1) \quad (9)$$

For simplicity, the value of L_1 and L_2 are chosen equally so that equal currents flow through these inductors. The output current I_{tune} from the analog adaptive neuron-fuzzy controller is used to switch the period of the PWM driving signal of the proposed scheme.

B. Hybrid batch(off-line) learning algorithm

Generally, when applying fuzzy control to a specific system, a control expert must set the approximate range of the membership functions. However, it is very difficult to extract the small changes in these membership functions. A training data set is required for an ANFIS to learn the proposed ABC-BES system. With this method, the membership functions can be modeled correctly. A hybrid learning rule based on least-square estimator (LSE) and gradient-descent method (GDM) is widely used [16]. A partition p of i -input is defined by giving an offset value a_i^p and distance d_i^p as illustrated in 6. Therefore, the total number of antecedent parameters to be trained with NP_i antecedents per input is $2 \times NP_i$. Concerning the consequent parameters, the number of crisp consequents to be adapted is equal to the number of rules, where the maximum is $m = \prod_{i=1}^n NP_i$ in which all possible rules are considered. The training data pair $(x_k; y_k')$, $k \in (1, 2, \dots, K)$ is used in this learning process. After the initial antecedent parameters and consequence parameters are identified, GDM will optimize the bell-shaped membership functions by computing the error function:

$$E = \frac{1}{2K} \sum_{k=1}^K (y_k - y_k')^2 \quad (10)$$

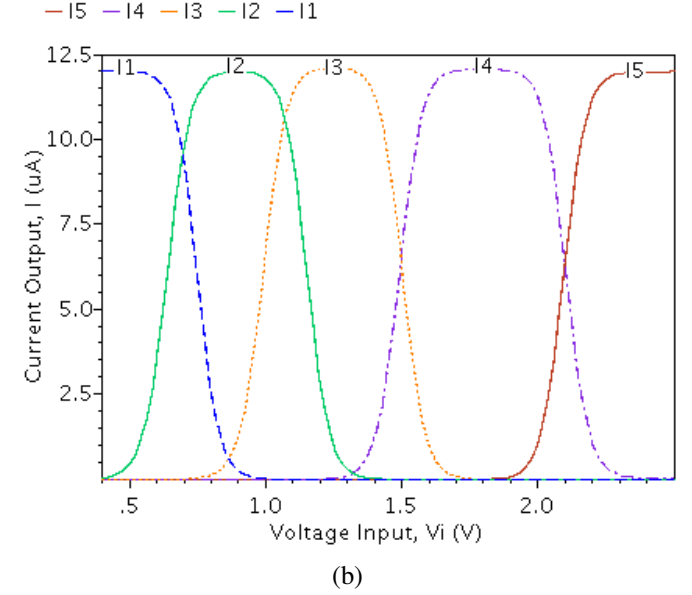
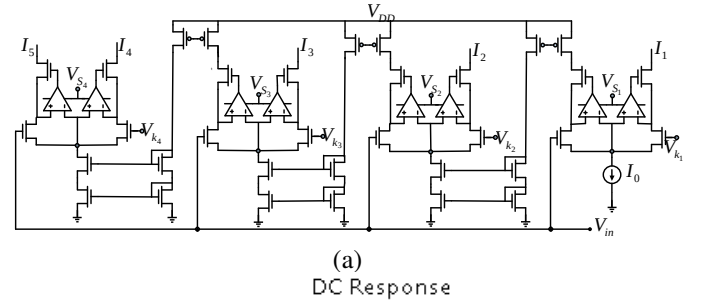


Fig. 7. Fuzzy membership function impact circuit. (a) Circuitry of membership functions with 5 labels. (b) Result of impact circuit currents.

Where: y_k is the actual output of the network for the k -training data, and y_k' is the desired output. Due to the use of this off-line learning mode, the learning process does not affect the voltage equalization time. The parameters are updated after all the training patterns are evaluated. The feedback line provides the actual stored outputs to the following training process. The off-line learning ANFIS algorithm is implemented in a microcontroller. In the proposed study, an Atmel microcontroller was utilized. Each training sample is collected by using sampling process. 10-bit ADC component in microcontroller works as voltage sensing part with the sampling frequency is 100 kHz. Sampling process is executed each 30 seconds, and takes 10ms for gathering 1000 samples in each. Therefore, the training process is executed every 5 minutes, after gathering 10,000 training samples.

C. Analog Adaptive Neuron-Fuzzy Controller (AANF)

The implementation of the proposed AANF controllers in a $0.13 - \mu\text{m}$ CMOS technology is presented in this section as follows:

- Layer 1:
The bell-shape membership functions are built around a regulated-cascode triode transconductor. For example, 7

shows the circuit for 5-label linguistic membership function. The slopes of the current curves are controlled with voltage V_S . The resulting membership function of V_B that has $V_{k1} = 0.65V$, $V_{k2} = 1.15V$, $V_{k3} = 1.65V$ and $V_{k4} = 2.15V$, respectively as demonstrated in 7(b). V_{B1} and voltage derivation D_e are the inputs for the proposed controller. The parameter set (a_i^p, d_i^p) of is updated from learning process in micro-processor as presented in the previous section.

$$a_i^p = \frac{V_{k_i}^{p-1} + V_{k_i}^p}{2}, \quad 2d_i^p = V_{k_i}^p - V_{k_i}^{p-1} \quad (11)$$

- Layer 2:

As described in 3, the layer 2 is designed by *minimum* circuit. A loser-take-all minimum *LTA-MIN* circuit is built up in this layer. This circuit choose minimum fuzzy current from layer-1 according the rules of $T - Norm$. This circuit is highly accurate, while achieving high current range without complexity. The circuit design and result are shown in 8.

- Layer 3,4,5 or defuzzification:

The final output after the neuron layer 5 is calculated as:

$$y = \frac{\sum_i \alpha_i s_i}{\sum_i \alpha_i} \quad (12)$$

The consequent parameter s_i depends on defuzzification strategy. In the proposed AANF controller, a current-mode D/A converter is used as digitally-programmable singleton s_i .

9 shows the proposed circuit for these layers operating in current-mode that have 5-bit resolution D/A converter.

The output current of this circuit is expressed as follows:

$$I_{out}^i = \beta_i s_i = \frac{[n \times (W/L)] \times I_i}{2^n} \quad (13)$$

$$I_{out} = \sum_{i=1}^m I_{out}^i = \sum_{i=1}^m \beta_i s_i \quad (14)$$

$$= \left(\sum_{i=1}^m (D_{n-1}^i 2^{n-1} + \dots + D_1^i 2^1 + D_0^i 2^0) \beta_i \right)$$

Where: m is number of rules. As aforementioned, the maximum number of rules is 25; with NP (number of membership functions per input) is 5. These values of singletons for the output current are determined after the ANFIS learning process. Meanwhile, this learning process forms the input parameters of the AANF defuzzification circuitry. The ratio of (W/L) of the lower NMOS device rows in 9(a) are equal to $1 : 2^n$. The total energy consumption by the AANF block is expressed in the following equation:

$$P \leq n(3m - 4)V_{DD}I_{0}^{\max} \quad (15)$$

While $V_{DD} = 2.5V$ and $I_0^{\max} = 12\mu A$. Therefore, the total loss power for AANF is less than $4.26mW$ which is insignificant in compare with the total power of system.

The overall BMS system with parameter and signal flows is

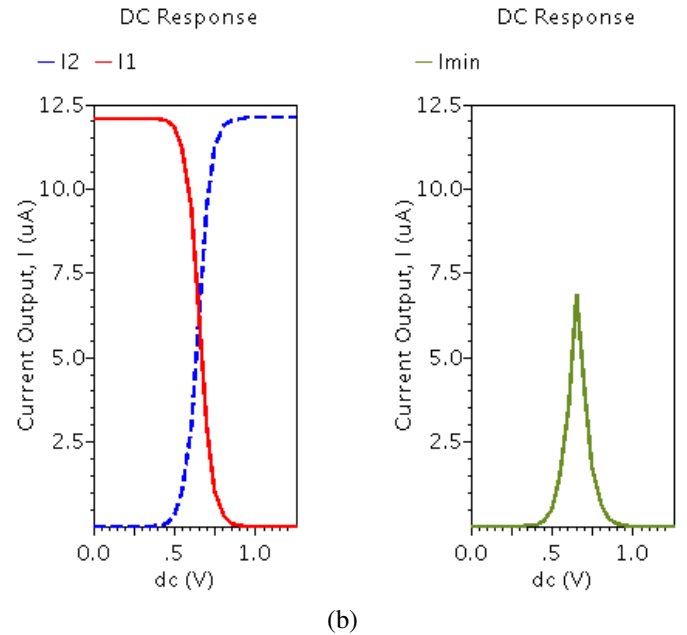
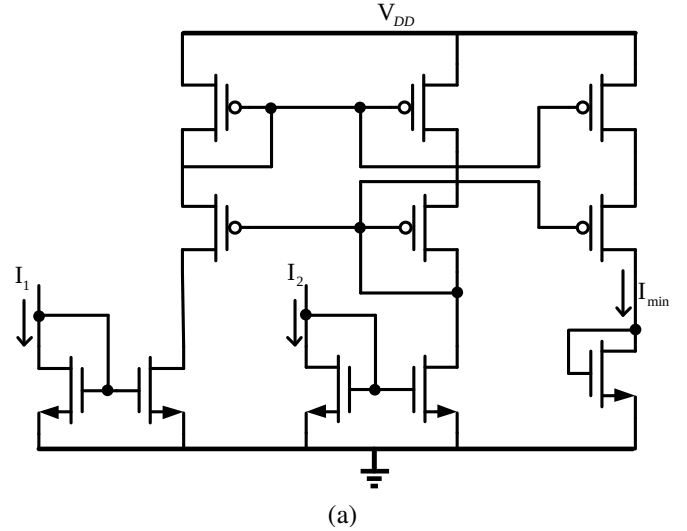


Fig. 8. (a) 2-input loser-take-all minimum circuit. (b) Result of LTA-MIN output current.

illustrated in 10. Thanks to the parallel and modular characteristic, only two battery cells with ICE are demonstrated. In the proposed system, the voltage sensor block is composed of voltage sensors for battery cells, ANFIS learning algorithm is constructed with a micro-processor, and the AANF block is designed in a $0.13-\mu m$ CMOS technology with power supply of $2.5V$.

IV. SYSTEM PERFORMANCE RESULTS

The advantage of the proposed system is about learning ability from various working conditions. Therefore, the battery cell voltages and the desired current to control PWM signal are evaluated within different conditions. Because the paramount importance of temperature in kinetic and transport phenomena of electrochemical systems, the battery voltage must account for thermal effects. The 11 shows the dependency of the battery cell voltages and the battery temperature. The wide

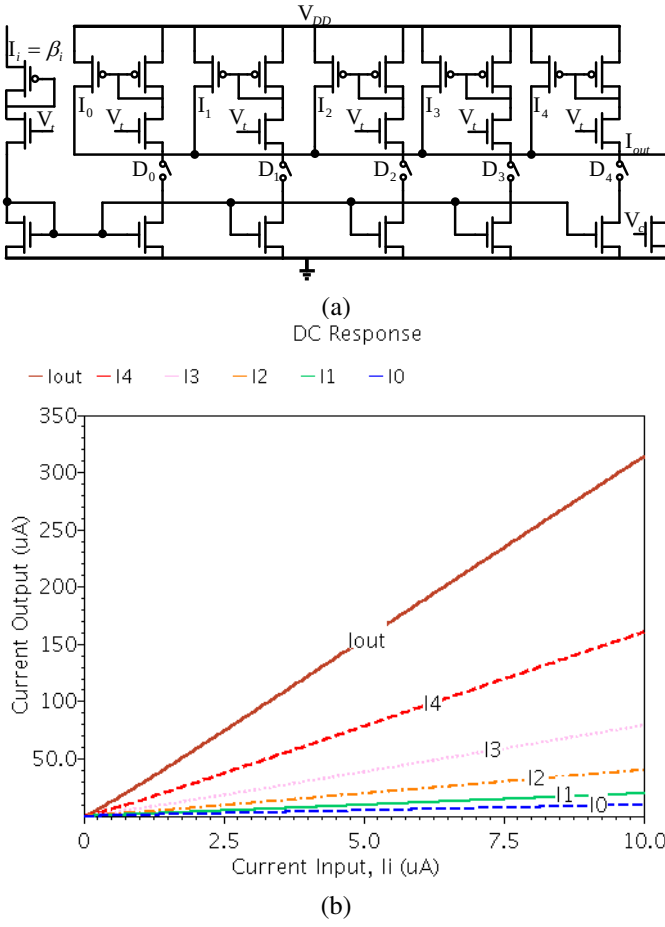


Fig. 9. (a) 5-bit resolution D/A defuzzification circuitry. (b) Result of defuzzifier inputs and outout current with input varies from 0-10uA, and $n = 32(D = 1111)$.

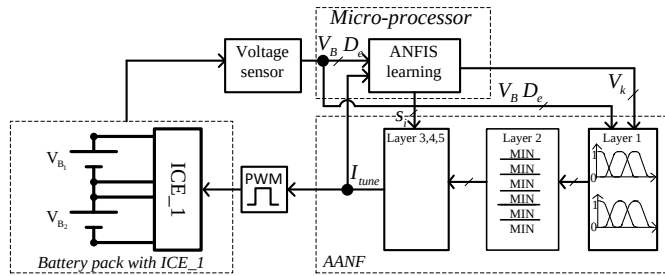


Fig. 10. Proposed ABC-BES with parameter and signal flows.

range of temperature [0-40] is considered to collect training data depended on temperature. As can be seen from this figure, the OCV of the battery cell is proportional to the working temperature [23]. Additionally, the stage-of-charge (SOC) is the ratio between the remaining or available capacity of a cell and the usable capacity which is affected by aging processes at electrodes [23]. The available capacity of the cell strongly depends on average discharge current, discharge time [23]. Therefore, it is obviously needed to evaluate the cell in workload conditions. 12 represents the cell voltage according the change of current in both charging and discharging states. For the first time of training process, the training samples are

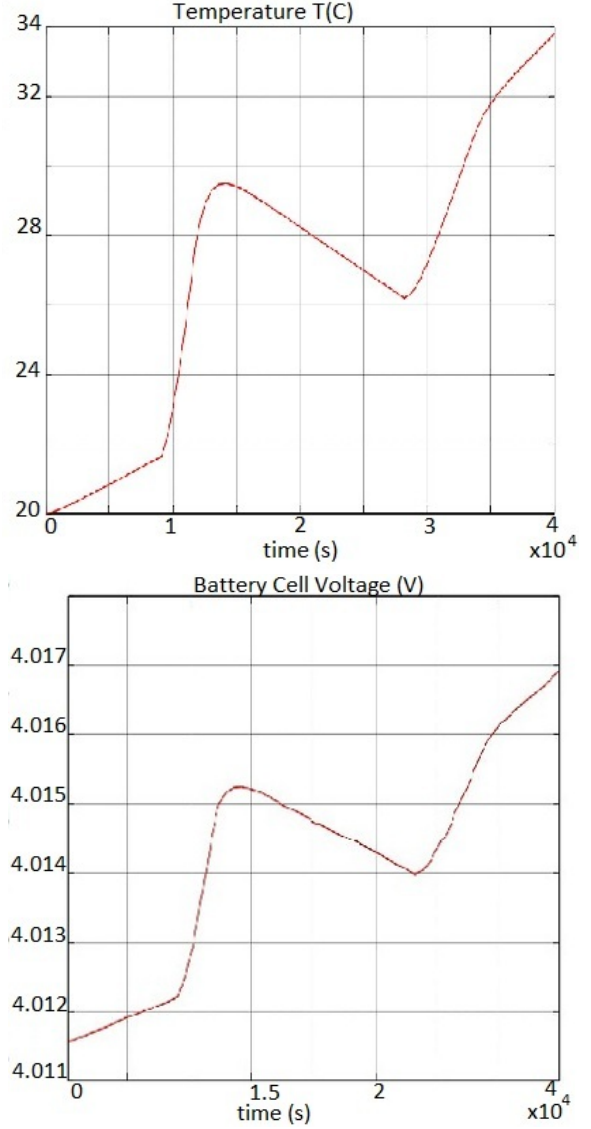


Fig. 11. The cell voltage versus temperature dependency.

collected from different temperatures and discharge/ charge currents to provide various working conditions. The training data set is composed of 10,000 pairs of input set $\{V_{B1}, D_e\}$ and current output I_{tune} .

For the experiment, 100 lithium-ion cells are serially connected as shown in 13. The battery cell used is the RHV-320065 model that has nominal voltage of 3.8V. The detail experimental conditions are summarized in I. 14 demonstrates

TABLE I
EXPERIMENTAL BATTERY SPECIFICATIONS

Specification	Battery Cell	Battery Pack
Capacity	16Ah	64Ah
Voltage	3.8V	$3.8 * 100(380V)$
Charging Current	Max 5C(80A)	Max 64A
Charging Voltage	$4.2 \pm 0.05V$	$4.2 * 100(420V)$
Charging Time	0.5C-4h	0.5C-4h
Operating Temperature	$0^{\circ}C \sim 45^{\circ}C$	$-20^{\circ}C \sim 60^{\circ}C$

the resulting voltage and current of inductor with initial

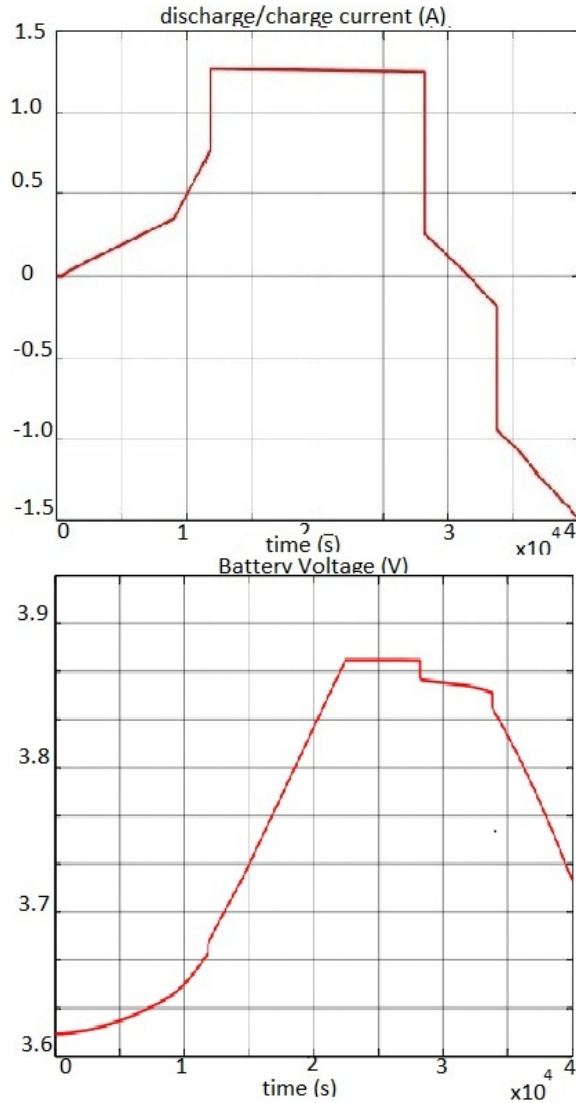


Fig. 12. The cell voltage versus battery current dependency.

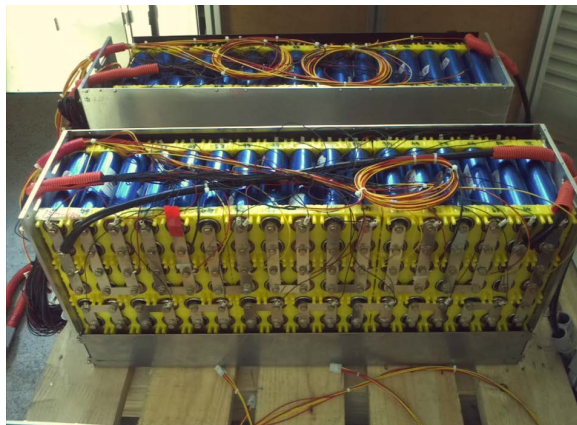


Fig. 13. Series connected lithium-ion battery strings.

voltages of $V_{B1} = 4.12V$, $V_{B2} = 3.65V$. The membership functions for each input are determined after the training process can be seen from 15. As illustrated in 15, the membership functions with adaptive parameters accurately elaborate the input voltages. As the result, the output current space is adaptive according to these parameters. The circuit parameters are. $L_1 = L_2 = 240\mu H$, $C = 430\mu F$. The switching frequency for initialization is selected at 50 kHz, with duty cycle of 50 percent. The control current surface generated by the ANFIS algorithm is shown in 16.

The fuzzy models of inputs are moderated automatically based on gathered training data. These parameters of fuzzy term for each input are transferred to AANF controllers. The experimental results demonstrate that the control currents generated from the proposed system have higher accuracy in comparison with previously reported BMS. The learning and adaptive ability of the proposed intelligent scheme provides advantages that could be expanded to various conditions and systems.

Finally, 17 demonstrates the error function evaluation in 10 of the proposed ABC-BES learning ability after 120 iterations. The error value of learning ability is superior to previously reported system with approximately $1.8 \times e(-5)$ after 120 iterations. In some previous researches about equalization battery cells using fuzzy logic theory to model current output which controls PWM signals [10]–[15], the membership functions for the equalization strategy using fuzzy logic controller are same as in the initialization while system is working at all the time. Therefore, the errors of fuzzy-only based systems are same as the proposed system at the initialization. In other words, the errors of the previous systems using fuzzy logic controllers are approximately $8 \times e(-5)$, as at the first iteration of the proposed system. While the working condition changes rapidly or sharply in term of temperature or current load, the error of the system without ANFIS controller will be accumulated to the control current. As long as the phenomenon continuously occurs, the much of error in output controller. Finally, it causes the malfunctions of the battery system or slows down the equalizing time because of lacking tracking and learning abilities.

V. CONCLUSION

This work successfully demonstrates an adaptive backward battery equalization system for Lithium-ion battery cell strings. An intelligent learning and adaptable algorithm of ANFIS is employed for tuning parameters of an analog adaptive neuro-fuzzy controller that is implemented in a standard $0.13 - \mu m$ CMOS technology. Based on these structures of the proposed system, hybrid learning ability and adaptability are enabled in battery equalization system. The performance of ANFIS was evaluated using standard error measurement. The proposed architecture achieves $1.8 \times e(-5)$ in term of learning error. The analog adaptive neuro-fuzzy controller is implemented with compact size for individual cell equalizer. The simulation and experimental results demonstrated the advantages of the proposed system.

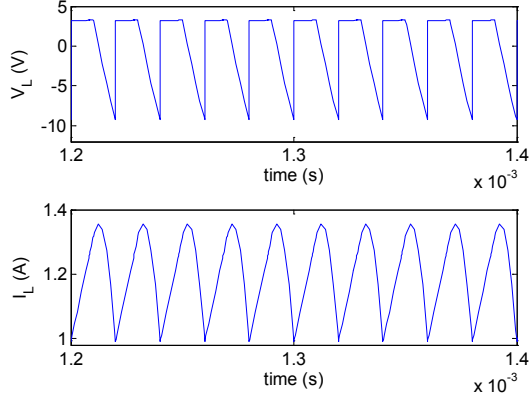


Fig. 14. L_1 's current and voltage experimental results for training set.

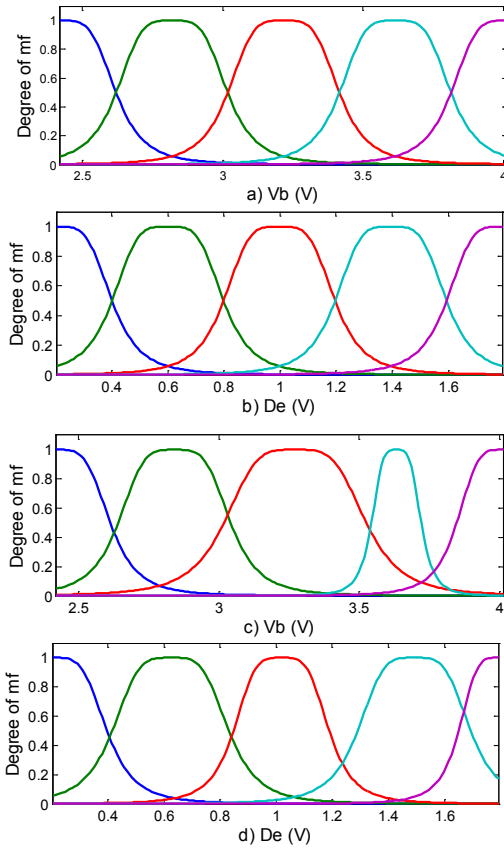


Fig. 15. Membership functions of input voltages before and after learning process with initial training datasets.

- Membership functions of input volated V_B before learning process.
- Membership functions of input volated D_e before learning process.
- Membership functions of input volated V_B after learning process.
- Membership functions of input volated D_e after learning process.

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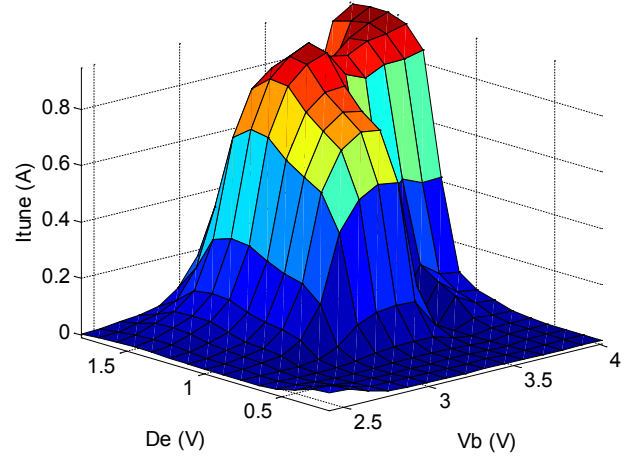


Fig. 16. Control current output I_{tune} model from learning process according to input voltages D_e and V_B .

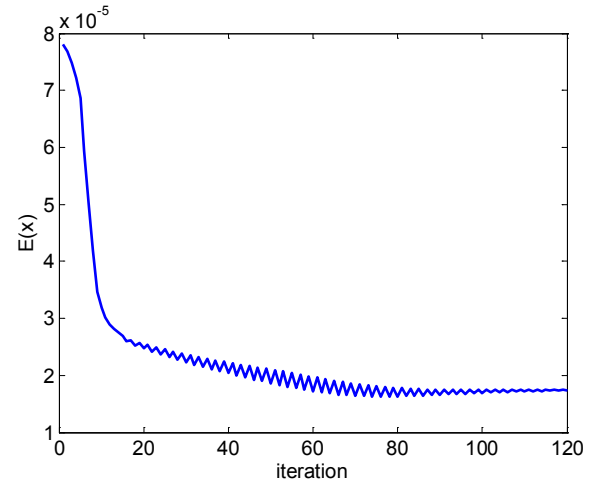


Fig. 17. Learning error in each iteration of the proposed equalization controller system.

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